Two Qubit Reduction sample with UCCSD fails with missing Z2_symmetrie

https://github.com/qiskit-community/qiskit-nature/issues/153

ROW 86

Two Qubit Reduction sample with UCCSD fails with missing Z2_symmetries argument #153

Closed

New issue

woodsp-ibm opened on Apr 23, 2021 · edited by woodsp-ibm

Edits

The following error results when running the code sample that follows. Run with `two_qubit_reduction=False` on the convertor and it works fine. I think the problem is more likely in the code in Terra, but since most of the sample is nature I raised it here but it can be transferred to Terra if it indeed turns out to be where the problem needs fixing.

```
  File "qiskit_nature/circuit/library/ansatzes/evolved_operator_ansatz.py", line 165, in <listcomp>
    evolved_ops = [self.evolution.convert((coeff * op).exp_i()) for op in self.operators]
  File "qiskit/opflow/operator_base.py", line 355, in __rmul__
    return self.mul(other)
  File "qiskit/opflow/primitive_ops/pauli_sum_op.py", line 130, in mul
    return super().mul(scalar)
  File "qiskit/opflow/primitive_ops/primitive_op.py", line 138, in mul
    return self.__class__(self.primitive, coeff=self.coeff * scalar)
TypeError: __init__() missing 1 required positional argument: 'z2_symmetries'
```

Here is the code sample to reproduce the above:

```
from qiskit_nature.drivers import PySCFDriver, UnitsType
from qiskit_nature.problems.second_quantization.electronic import ElectronicStructureProblem
# Use PySCF, a classical computational chemistry software
# package, to compute the one-body and two-body integrals in
# electronic-orbital basis, necessary to form the Fermionic operator
driver = PySCFDriver(atom='H .0 .0 .0; H .0 .0 0.735',
                     unit=UnitsType.ANGSTROM,
                     basis='sto3g')
problem = ElectronicStructureProblem(driver)
# generate the second-quantized operators
second_q_ops = problem.second_q_ops()
main_op = second_q_ops[0]
num_particles = (problem.molecule_data_transformed.num_alpha,
                 problem.molecule_data_transformed.num_beta)
num_spin_orbitals = 2 * problem.molecule_data.num_molecular_orbitals
# setup the classical optimizer for VQE
from qiskit.algorithms.optimizers import L_BFGS_B
optimizer = L_BFGS_B()
# setup the mapper and qubit converter
from qiskit_nature.mappers.second_quantization import ParityMapper
from qiskit_nature.converters.second_quantization import QubitConverter
mapper = ParityMapper()
converter = QubitConverter(mapper=mapper, two_qubit_reduction=True)
# map to qubit operators
qubit_op = converter.convert(main_op, num_particles=num_particles)
# setup the initial state for the ansatz
from qiskit_nature.circuit.library import HartreeFock
init_state = HartreeFock(num_spin_orbitals, num_particles, converter)
# setup the ansatz for VQE
from qiskit_nature.circuit.library import UCCSD
ansatz = UCCSD(converter,num_particles,num_spin_orbitals,1,init_state)
from qiskit import Aer
backend = Aer.get_backend('statevector_simulator')
# setup and run VQE
from qiskit.algorithms import VQE
algorithm = VQE(ansatz,
                optimizer=optimizer,
                quantum_instance=backend)
result = algorithm.compute_minimum_eigenvalue(qubit_op)
print(result.eigenvalue.real)
```

The issue was originally raised on the external Slack channel:
https://qiskit.slack.com/archives/CB6C24TPB/p1619192806297800

woodsp-ibm added type: bug on Apr 23, 2021

woodsp-ibm assigned ikkoham on Apr 23, 2021

Assignees
ikkoham

Labels
type: bug

Type
No type

Projects
No projects

Milestone
No milestone

Relationships
None yet

Development
No branches or pull requests

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Participants

```
# setup the mapper and qubit converter
from qiskit_nature.mappers.second_quantization import ParityMapper
from qiskit_nature.converters.second_quantization import QubitConverter
mapper = ParityMapper()
converter = QubitConverter(mapper=mapper, two_qubit_reduction=True)
# map to qubit operators
qubit_op = converter.convert(main_op, num_particles=num_particles)
# setup the initial state for the ansatz
from qiskit_nature.circuit.library import HartreeFock
init_state = HartreeFock(num_spin_orbitals, num_particles, converter)
# setup the ansatz for VQE
from qiskit_nature.circuit.library import UCCSD
ansatz = UCCSD(converter,num_particles,num_spin_orbitals,1,init_state)
from qiskit import Aer
backend = Aer.get_backend('statevector_simulator')
# setup and run VQE
from qiskit.algorithms import VQE
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woodsp-ibm assigned ikkoham on Apr 23, 2021

mrossinek on Apr 23, 2021 Member

Seems to be a duplicate of this issue: Qiskit/qiskit#6127

mrossinek on Apr 28, 2021 Member

Once Qiskit/qiskit#6315 is merged we need to remove this line in order to close this issue.
While validating the fix of the PR in Terra I noticed that we can probably also remove the `@slow_test` decorators because these tests aren't actually that slow..

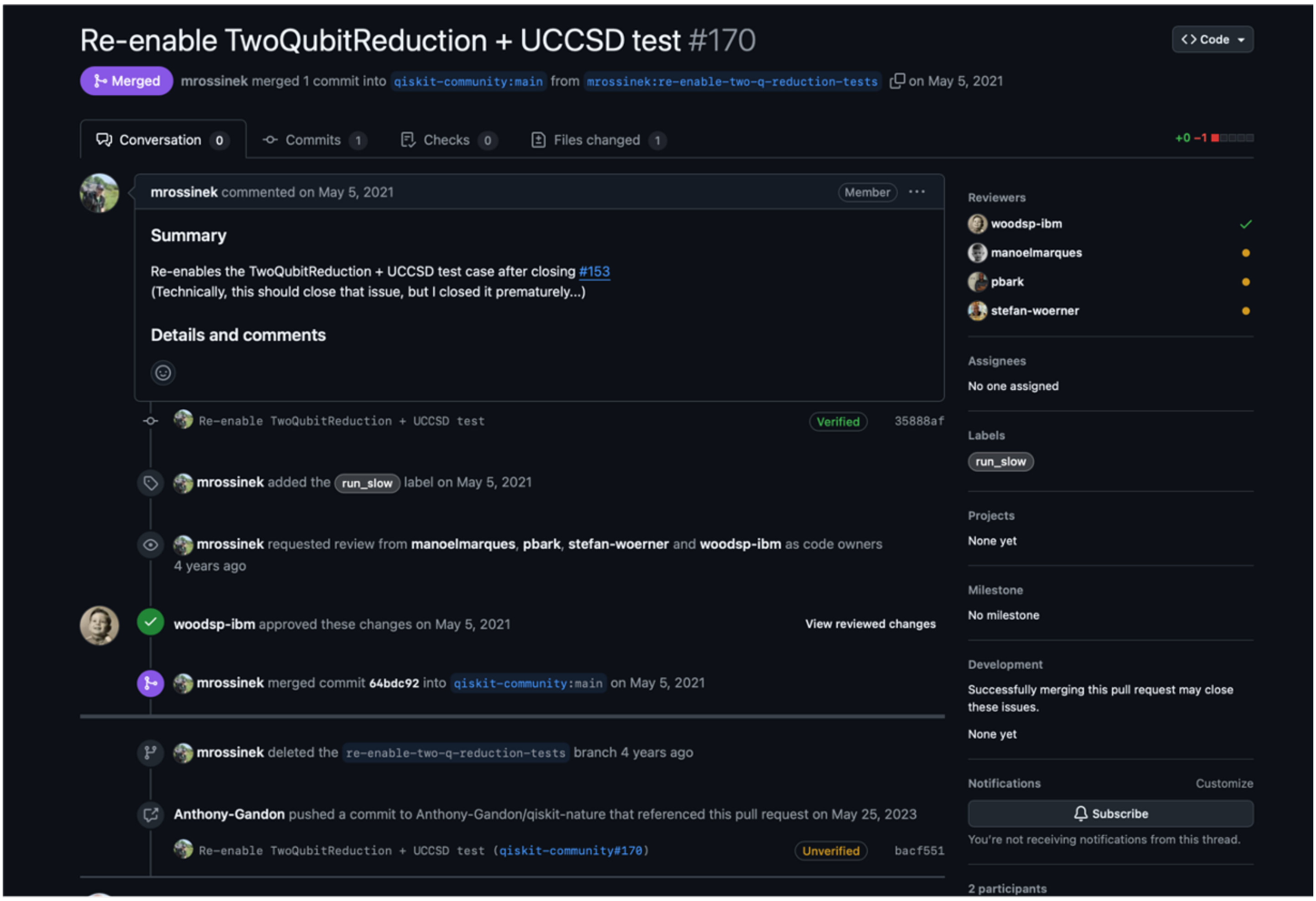
mrossinek on May 5, 2021 Member

Closed by Qiskit/qiskit#6315

mrossinek closed this as completed on May 5, 2021

mrossinek mentioned this on May 5, 2021

Re-enable TwoQubitReduction + UCCSD test #170



Re-enable TwoQubitReduction + UCCSD test #170

Merged mrossinek merged 1 commit into `qiskit-community:main` from `mrossinek:re-enable-two-q-reduction-tests` on May 5, 2021

Conversation 0 | Commits 1 | Checks 0 | Files changed 1 | +0 −1

mrossinek commented on May 5, 2021 — Member

Summary

Re-enables the TwoQubitReduction + UCCSD test case after closing #153
(Technically, this should close that issue, but I closed it prematurely...)

Details and comments

`Re-enable TwoQubitReduction + UCCSD test` — Verified — 35888af

mrossinek added the run_slow label on May 5, 2021

mrossinek requested review from **manoelmarques**, **pbark**, **stefan-woerner** and **woodsp-ibm** as code owners
4 years ago

woodsp-ibm approved these changes on May 5, 2021 — View reviewed changes

mrossinek merged commit **64bdc92** into `qiskit-community:main` on May 5, 2021

mrossinek deleted the `re-enable-two-q-reduction-tests` branch 4 years ago

Anthony-Gandon pushed a commit to Anthony-Gandon/qiskit-nature that referenced this pull request on May 25, 2023

`Re-enable TwoQubitReduction + UCCSD test (qiskit-community#170)` — Unverified — bacf551

Reviewers
woodsp-ibm
manoelmarques
pbark
stefan-woerner

Assignees
No one assigned

Labels
run_slow

Projects
None yet

Milestone
No milestone

Development
Successfully merging this pull request may close these issues.
None yet

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2 participants